25900236 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBIF200 - SEQUENCE INFORMATICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. State sanger sequencing method? [U] / [CO1]
2. How does the choice of BLAST variant depend on whether the input sequence is nucleotide or protein? [U] / [CO2]
3. Indicate why is protein family identification important? [A] / [CO4]
4. How can multiple sequence alignment assist in the annotation of newly sequenced genomes? [U] / [CO2]
5. Give two examples of widely used tools for pairwise alignment. [K] / [CO2, CO3]
6. Define main purpose of a primer in sequencing? [U] / [CO1]
7. Analyze why PSI–BLAST is considered more powerful for protein family detection compared to PHI–BLAST. [An] / [CO3]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

8.	Identify one biological application where the Needleman–Wunsch algorithm is more suitable than local alignment methods.	[U] / [CO2]
9.	Discuss how dot plots help in identifying repeated sequences and palindromic regions.	[U] / [CO2, CO3]
10.	Define the tools used for MSA.	[U] / [CO2]
11.	Describe how a scoring matrix influences alignment results?	[U] / [CO2]
12.	Compare sequence profile with regular expression in detecting motif.	[An] / [CO3]
13.	Explain the advantage of profile in studying conserved regions.	[U] / [CO2]

[5x4 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

14.	Comprehend the significance of "Blocks" in BLOSUM matrices.	[U] / [CO2]
15.	Explain how multiple sequence alignment is used in evolutionary studies and species identification.	[A] / [CO4]
16.	Explain the principle and steps of Maxam-Gilbert sequencing.	[U] / [CO1]
17.	Describe the importance of functional annotation in genome projects.	[A] / [CO5]

[10x2 = 20]